

LAB 11 — TESTING & DEBUGGING IN FLUTTER (TASKLY APP)

Module 11 — Flutter Mobile Development

Total Tasks: 5 (4 exercises + 1 final consolidation)

Total Grade: 100%

SECTION A — Learning Objectives

After completing Lab 11, students will be able to:

- Write **unit tests** for pure Dart logic (Task model, TaskRepository).
 - Write **widget tests** to validate UI behaviors in Flutter.
 - Perform **navigation tests** to ensure screens transition correctly.
 - Perform **integration testing** to validate end-to-end flows.
 - Use **Flutter DevTools** to inspect widget trees, constraints, rebuilds, and performance.
 - Combine multiple test types into one complete **Flutter test suite**.
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SECTION B — LAB INSTRUCTIONS

LAB 11.1 — Unit Testing: Task Model & Repository

Description

Implement unit tests to validate logic in the Task model and TaskRepository.

Tasks

1. Create folder: test/unit/
2. Create file: task_model_test.dart
 - Test default completed value
 - Test toggle() switches true ↔ false
3. Create file: task_repository_test.dart
 - Test addTask()

- Test deleteTask()
 - Test updateTask()
4. Use the **AAA pattern** (Arrange → Act → Assert).
 5. Fix any logic errors discovered by tests.

Deliverables

- test/unit/task_model_test.dart
 - test/unit/task_repository_test.dart
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LAB 11.2 — Widget Testing: Task List UI

Description

Write widget tests to validate UI rendering and task addition in TaskListScreen.

Tasks

1. Create file:
test/widget/task_list_widget_test.dart
2. Write tests for:
 - **Empty State:** “No tasks yet. Add one!”
 - **Add Task:** enter text → tap add → verify UI updates
 - **Multiple Tasks:** add two tasks → verify both visible
3. Use pumpWidget(), enterText(), tap(), pump().

Deliverables

- test/widget/task_list_widget_test.dart
-

LAB 11.3 — Navigation Testing: Task List → Task Detail

Description

Write tests to verify navigation to TaskDetailScreen.

Tasks

1. Create file:
test/widget/task_navigation_test.dart
2. Seed repository with at least one task.
3. Pump TaskListScreen.
4. Tap seeded task → call pumpAndSettle().
5. Validate:
 - AppBar title: “Task Detail”
 - TextField key: detailTitleField

Deliverables

- test/widget/task_navigation_test.dart
-

LAB 11.4 — Integration Testing & DevTools Debugging

Description

Perform a full flow integration test and inspect UI/performance using DevTools.

Part A — Integration Test

Tasks

1. Create file:
test/widget/task_integration_test.dart
2. Simulate full flow:
 - Add “Original title”
 - Tap task → open detail
 - Edit → “Updated title”
 - Save
 - Verify updated title appears in list

Deliverables

- test/widget/task_integration_test.dart

Part B — DevTools Debugging

Tasks

1. Run Taskly in debug mode
2. Open DevTools:
 - **Widget Inspector**
 - **Performance Timeline**
3. Capture screenshots:
 - Widget tree
 - Performance trace
4. Write a short report (½–1 page):
 - One potential layout issue
 - One potential performance issue
 - How tests or DevTools help detect them

Deliverables

- DevTools screenshots (2)
- Debugging report (PDF/DOC)

LAB 11.5 — Final Consolidation: Complete Test Suite

Description

Combine all tests into a complete structured test suite.

Tasks

1. Organize folders:

test/

└─ unit/

└─ widget/

└─ integration/

2. Ensure presence of:

- Unit tests
- Widget tests
- Navigation tests
- Integration tests

3. Run:

flutter test

4. Fix all failing tests.

5. Write a **Test Summary Document**:

- number of tests
- types of tests
- behaviors validated
- known limitations

Deliverables

- Full test suite
- Test summary PDF

SECTION C — SUBMISSION REQUIREMENTS

Students must submit:

1. Source Code ZIP

Must include:

- /lib

- /test
- DevTools screenshots
- Reports

2. Required Screenshots

- Widget Inspector
- Performance Timeline

3. Reports

- Debugging Report (Lab 11.4)
- Test Suite Summary (Lab 11.5)

4. Naming Convention

Module11_Testing_<StudentID>_<FullName>.zip

SECTION D — GRADING RUBRIC (100%)

Component	Weight
LAB 11.1 – Unit Testing	20%
LAB 11.2 – Widget Testing	20%
LAB 11.3 – Navigation Testing	20%
LAB 11.4 – Integration Testing + DevTools Debugging	30%
LAB 11.5 – Full Test Suite Consolidation	10%
TOTAL	100%